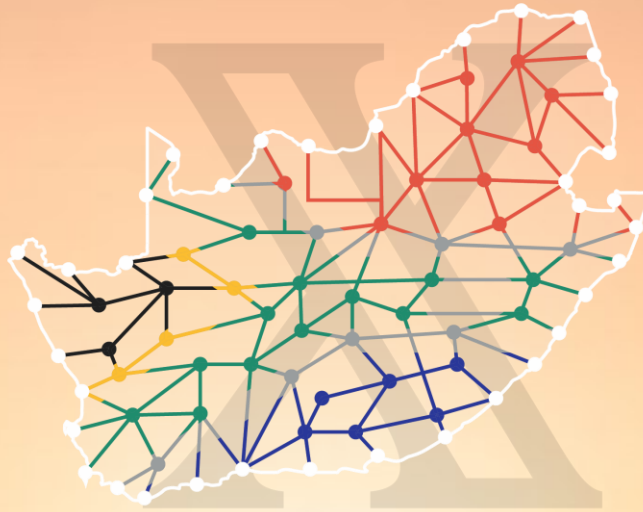




Deep Learning Indaba **X**
South Africa

2021 Report



3 September to 26 November 2021

A virtual roadshow of 13 groups from across the
South African machine learning landscape

WHAT IS AN INDABA $\times$?

DEEP
LEARNING
INDABA



The Indaba $\times$ South Africa is one of many across the continent

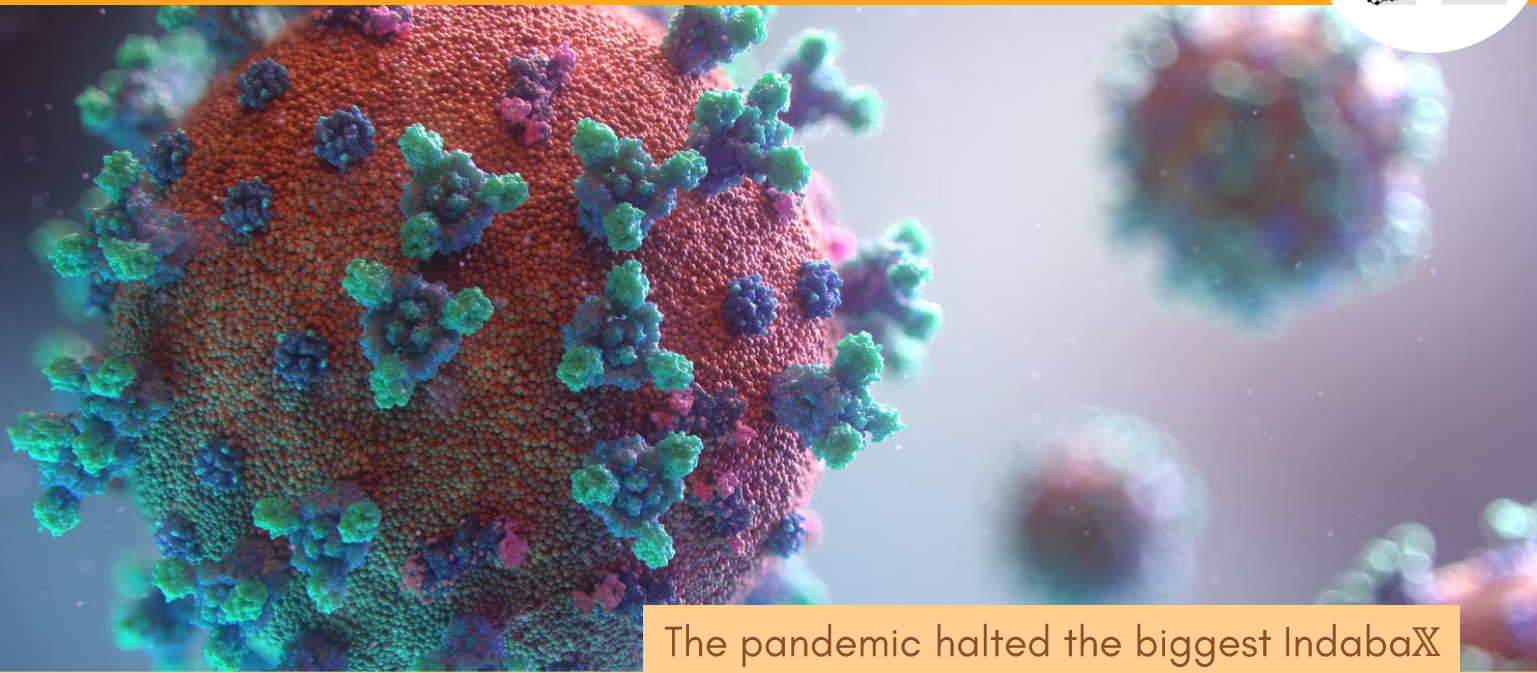
The world continues to change at a rapid pace, with artificial intelligence and machine learning increasingly playing an integral role in human advancements. These technologies are expected to shift power dynamics and shape the future in unprecedented ways in what is termed the "4th industrial revolution". Africa, in turn, is primed to leverage an ever-growing human capacity in order to make important contributions in the machine learning community.

Born from the desire to diversify this community defining the future, the pan-African Deep Learning Indaba was founded in 2017 to spread knowledge and strengthen African capacity in machine learning. Seeded from the root conference, a diaspora of 27 countries across the African continent have hosted an Deep Learning Indaba $\times$ event since 2018, each a community-driven meeting disseminating knowledge wider and deeper. The South African Indaba $\times$ aims to be the pinnacle education and research-focused artificial intelligence conference in the country, having hosted an event each year since 2018 and reaching more than 500 delegates.

In building capacity for machine learning in South Africa, we seek to create lasting connections, drive innovative developments, and proactively influence how recent advances shape our society. The Deep Learning Indaba $\times$ South Africa therefore presents an opportunity for students, researchers, and industry to come together from around the country to learn, share, and grow machine learning in South Africa and beyond.

More so than ever, the Indaba $\times$ means to experiment with innovative ways of knowledge sharing. For 2021, this meant a roadshow of academic and industry groups across the country sharing who they are and what they are pursuing.

INDABAX 2020 AND THE GLOBAL PANDEMIC



The pandemic halted the biggest IndabaX

We live in an amazing time with access to so many resources, knowledge, and opportunities available to us. Building a community is something special and it takes lots of people putting in a lot of effort. That is why it was particularly hard to postpone the Deep Learning IndabaX South Africa 2020 conference for the 300 people planning to attend. As a team, we tried our best to maintain a level-headed, rational approach to organising a great conference amidst the very real concern of the COVID-19 pandemic (severe acute respiratory syndrome coronavirus 2 - SARS-CoV-2). In fact, we sent our email off 2-3 days before the SA government officially suspended large gatherings. We took the proactive approach to make sure our message was clear: we care about our community.

We felt it was in the best interest of the community to not put unnecessary risk on delegates, invited speakers, sponsors, and organisers. Most of our international speakers had travel restricted by either their country or company. Many conferences, local and abroad, have been cancelled due to fears. We thought it would be irresponsible to continue organising a gathering of 300 people when fear and worry were at the forefront of our minds. We envisioned a conference of friendly laughter, thoughtful insights, and warm embraces. To continue at that stage would not meet this vision.

Along with the biggest and longest IndabaX ever planned, we had the largest team of passionate organisers than ever. Special mention to Naomi Molefe, Raesetje Sefala, Vongani Maluleke, Windy Mokuwe, and Vukosi Marivate for being integral leaders of the IndabaX 2020: we are sorry you could not see the fruits of your labour.



Naomi Molefe



Raesetje Sefala



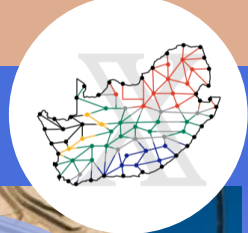
Vongani Maluleke



Windy Mokuwe



Vukosi Marivate



INDABAX 2021: ROADSHOW

Roadshow

13 hosts
13 events
30+ hours
50+ speakers

Community Forum

416 members
1000+ messages
4 server followings
604 YouTube subscribers

Competition

2 hackathons
4 roadshow rewards
30 data vouchers
R100k+ given away

The core spheres of the 2021 edition

The COVID-19 pandemic forced us to re-evaluate how we learn and connect with each other. Although we were individually isolated, we had the opportunity to digitally bring down barriers and strengthen the South African machine learning community.

Traditionally, the IndabaX prides itself on the focus it gives to the human connections formed through an intense conference journey. We do not believe that a direct translation to an online conference would replicate the magic, and therefore did not even attempt to do so. But we still wanted to share knowledge, open opportunities, and connect people in the most interactive manner possible. This is why for 2022, **the community itself was the main contributor to the IndabaX!**

Roadshow

The roadshow takes place from 3 September - 27 November and offers a **series of events, each of which is hosted by a different team of local organisers and/or institutions.**

The local teams design the content of the event in collaboration with the main IndabaX organising committee. The events can take the form of a lab showcase, hackathon, industry panel, student talks, ML pub quiz, or any other engaging, fun and informative activity.

Community Forum

The growth of a decentralised IndabaX South Africa community forum, where everyone can sign up and interact with each other.

The space is a safe peer-to-peer platform for aspiring and seasoned South African machine learning enthusiasts to exchange knowledge, search for help, recruit talent and build an inclusive culture around data and ML.

Competition

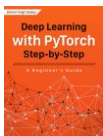
The goal of any competition is to win. However, the purpose of a competition is to **push ourselves** and each other to innovate further, achieve new levels of insights, and create opportunities for both individual and collective achievement.

Here, the IndabaX seeks to bring out the **best and brightest minds** in machine learning within South Africa to share their achievements, community engagements, and to build solutions to an industry problem (in the form of a hackathon).



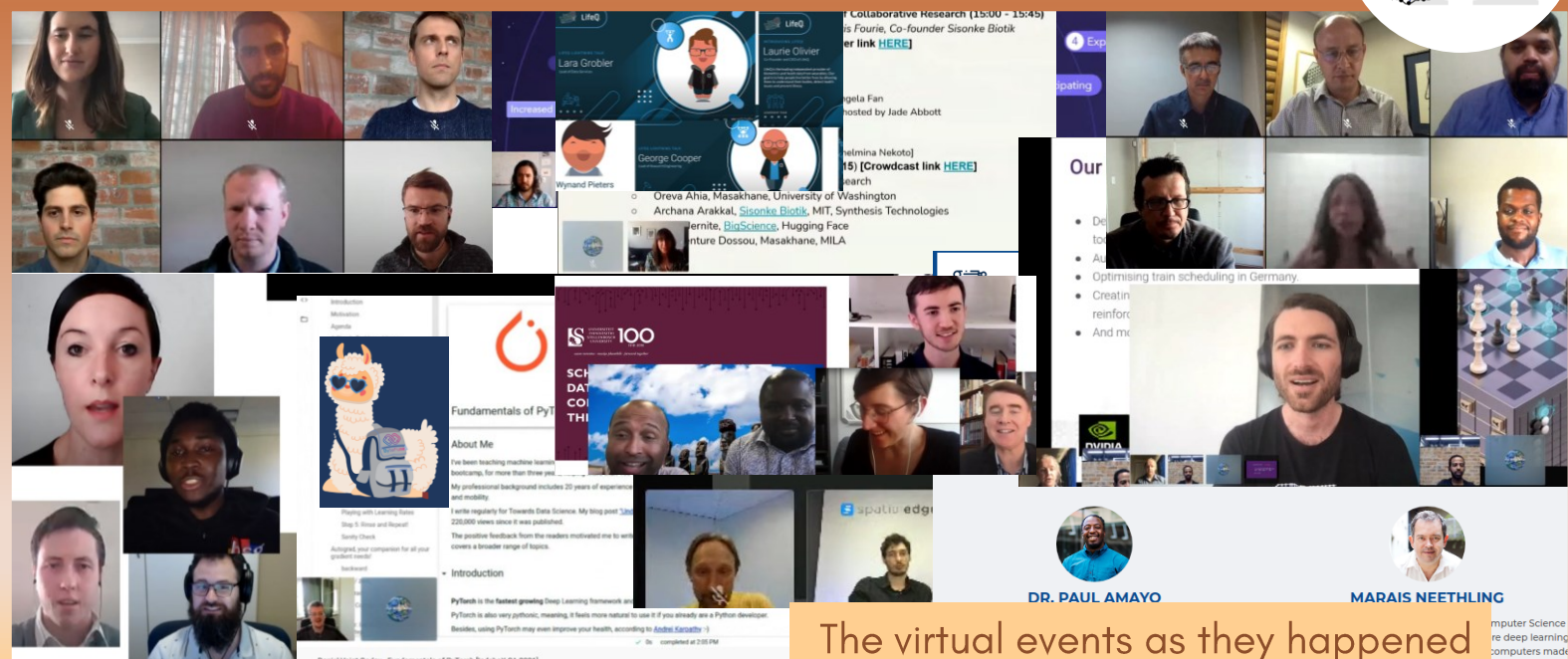
ROADSHOW SCHEDULE



3rd Sept		✕ How to turn a Machine Learning Use Case into a Successful Startup
7th Sept		✕ AI Expo Africa 2021
10th Sept		✕ Grassroots AI: The unreasonable effectiveness of collaborative research
17th Sept		✕ Tuts, a Talk and AGI !!
1st Oct		✕ Fundamentals of PyTorch: Building a Model Step-by-Step
8th Oct		✕ Tutorial on Notebook workflows for reproducible data science
15th Oct		✕ AI Night with DataProphet



ROADSHOW SCHEDULE cont.



The virtual events as they happened

22nd
Oct

X ByteFuse: Search and optimisations with llamas

29th
Oct

X Playing StarCraft and saving the world using multi-agent reinforcement learning!

5th
Nov

X Career Day: opportunities for students, graduates and those wanting to change career

12th
Nov

X AI UPtake: Panel discussion on collaborative research

19th
Nov

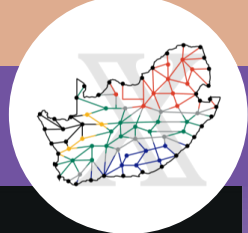
X From sensors to health data analysis

26th
Nov

X Roadshow highlights, awards, and what's next



Most videos available on our YouTube channel:
<https://www.youtube.com/c/DeepLearningIndabaX>



ROADSHOW REWARD WINNERS

Hackathon Heroes

R20 000 | Cape AI Hackathon powered by Zindi

Do you have what it takes to predict whether or not a customer is about to take an item from a shelf?

The Cape AI team is pleased to announce that they are partnering with Zindi on their latest hackathon for IndabaX. Test your ability to read shoppers and be rewarded with a prize from our R20 000 prize pot. The Cape AI x IndabaX Hackathon is all about predicting whether or not a stream of 30 video frames from a real customer's shopping session contains a "take" action.

1. **E&Ez**
2. **Johan010**

R18 000 | LifeQ Hackathon powered by Zindi

Can you detect dangerously irregular heart beats (arrhythmias) from a person's wearable data?

LifeQ is partnering with Zindi for an IndabaX hackathon! Keeping an eye on heart health is important.

The objective of the challenge is to build a predictive model that accurately classifies the presence of chronic atrial fibrillation or other arrhythmias given the continuous time series data of intervals between a person's heart beats.

1. **Matthew Baas**
2. **Christopher Cleghorn**
3. **Tiffany Woodley**

Two unique hackathons hosted on Zindi allowed participants to show off their skills and win cash

Cooldest coder

R10 000

rewards outstanding code-based projects or contributions to open-source code; an emphasis lies on best practices in software development and innovative solutions

Nathan Michlo

🌐 Disent – A modular disentangled representation learning framework built with PyTorch Lightning

Most Rigorous Researcher

R10 000

rewards outstanding contributions or a single outstanding contribution to machine learning research; a focus on impact for South Africa will be considered favourably

Raesetje Sefala

🌐 Constructing a Visual Dataset to Study the Effects of Spatial Apartheid in South Africa. Raesetje Sefala, Timnit Gebru, Luzango Mfupe, Nyalleng Moorosi, Richard Klein. NeurIPS 2021.

InstaDeep's Extraordinary Engagement

R10 000

rewards extraordinary efforts in strengthening and growing the South African machine learning community

Jade Abbott

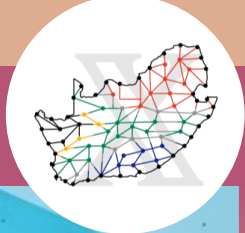
🌐 Masakhane – A grassroots NLP community for Africa, by Africans

Persistent Passion

R10 000

this is an open category for anyone who does not fit the above categories, but still deserves to be acknowledged for their diligent pursuit of machine learning, especially under adverse circumstances

Ashley Ehlers



HOSTS AND SPONSORS



Without the support of our amazing and generous sponsors, the Deep Learning IndabaX South Africa 2021 would not have been able to reach its aims and dreams.

Special thanks to Deep Learning Indaba, for continuing to push machine learning in Africa and pull communities together; World Wide Science, for sharing and hosting most of our events for free on Crowdcast; and Venues for Conferences in Africa, for assisting with our accounting while we registered as a not-for-profit company.



DEEP LEARNING
INDABA

ASHANTI |
DATA SCIENCE SPECIALIST



▶ InstaDeep™

dataprophet
machine learning specialists

CAPEAi



LifeQ



MaSS
DST-NRF CENTRE OF
EXCELLENCE IN
MATHEMATICAL &
STATISTICAL SCIENCES

ZIND!

moon
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UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Google
alis
AI Driven Investments

IBM

NITheCS

National Institute for
Theoretical and
Computational Sciences



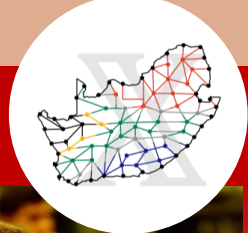
AI EXPO
AFRICA 2021
AUTOMATION & INTELLIGENCE FOR THE FUTURE



spatialedge

Venues
FOR CONFERENCES
IN AFRICA





ORGANISATION



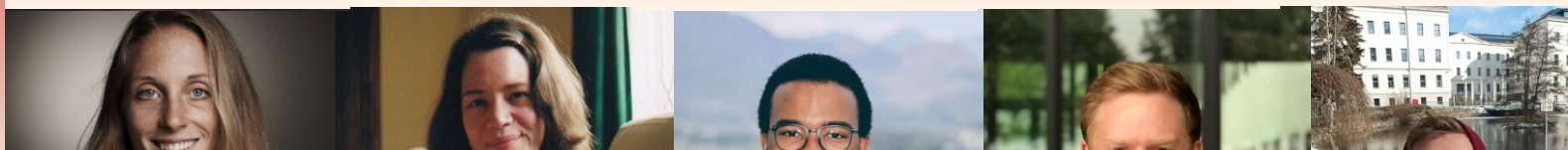
Siobhan Hall

Avashlin Moodley

Jeanne Daniel

Chris Fourie

Christopher Cleghorn



Maria Schuld

Anna Bosman

Sicelukwande Zane

Devin
Taylor

The 2022 team

Christopher
Currin

A group of highly motivated team players from across the country rose to the occasion of organising one of the biggest IndabaX events on the continent and one of the biggest artificial intelligence and machine learning event series in the country.

While spread across the country, and with some having never met in real life, the modern age has meant remote working is true for conference organisation too. It helped using a central repository of information, having leaders clearly documenting and communicating decisions on a Discord server. Because of the year-long endeavour, many organisers were awarded new professional positions and hence have multiple affiliations reflecting the international expertise of our South African organisers.

This year, the **IndabaX officially became a non-profit company (NPC)**, registered under the name "Deep Learning IndabaX South Africa", with directors Christopher Currin, Maria Schuld, and Anna Bosman. This formality reflects our ongoing commitment to growing machine learning in South Africa.

LIST OF ORGANISERS

SIOBHAN HALL

Stellenbosch University
University of Oxford

JEANNE DANIEL

CapeAI
Stubben Edge

AVASHLIN MOODLEY

Entelect
Vodacom

SICELUKWANDE ZWANE

University of the Witwatersrand
University College London

CHRIS FOURIE

University of the Witwatersrand
LifeQ

CHRISTOPHER CLEGHORN

University of the Witwatersrand

DEVIN TAYLOR

Nedbank
WovenLight

ANNA BOSMAN

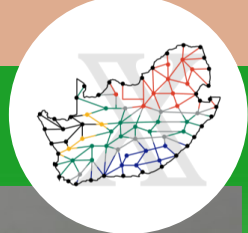
University of Pretoria

MARIA SCHULD

University of KwaZulu-Natal
Xanadu

CHRISTOPHER CURRIN

University of Cape Town
Institute of Science and Technology Austria



BUDGET



Although we have fewer expenses, the opportunity to celebrate our community with substantial monetary prizes was made possible with the support of our sponsors, led by the Deep Learning Indaba.

For 2022 and beyond, the IndabaX has become an official node of the DST-NRF Centre of Excellence for Mathematical and Statistical Sciences (CoE-MaSS) for continued institutional support.

EXPENSES	
Item	Cost
Roadshow rewards prizes	4 x R10,000
Data vouchers giveaway and runner-up prizes	R30,000
Zindi + Cape AI Hackathon prizes	R15,000
Accounting*	R7,000
TOTAL	R92,000

*Estimate, since some expenses are only due in 2022

SPONSORSHIPS (excl. prizes)	
Partner	Amount
Deep Learning Indaba	\$2,000 (R27,697.72)
InstaDeep	\$1,400 (R20,766.52)
SpatialEdge	R7,500
LifeQ	R3,750 + hackathon
Cape AI	R5,000 + hackathon
AI Expo Africa	4x tickets
Zindi	Hackathon hosting
NITheCS	30 x R1,000 data vouchers
Venues for Conferences in Africa	Logistical support
World Wide Science	Hosting infrastructure
TOTAL	R94,714.24

The surplus will fully contribute towards the IndabaX 2022 budget.

TOTAL

R2,714.24



CONCLUSIONS



With the rapid advancement of AI across the world, it is imperative that Africa not only has a voice, but an authority in the field. One way to stand out is by embracing “African problems” through our domain knowledge and unique contexts to solve local problems that impact global solutions. Doing it alone is never an option. The (South) African AI community is growing at an exponential rate year-on-year. The connections formed at these events form a network of people integral to growing the field and each other.

Collaborations between local and international universities and companies are vital to growth and development. In addition, the noticeable number of international African students at the Deep Learning Indaba $\mathbb{X}$ shows hunger for greater partnerships on the continent. Still, inter-country policies need to be addressed to facilitate better communication between collaborators. Fortunately, the Deep Learning Indaba movement is a means to strengthen African machine learning through knowledge transfer and connection building.

These events last beyond the schedule and reach beyond the attendees. The passion of everyone involved creates power. Powerful algorithms. Powerful connections. Powerful people that grow the space and seed the next event.

We truly appreciate everyone involved in creating a special event. From those attending defining the experience to the sponsors supporting the organisers in executing their vision. We are thrilled to see so many people eager to push what is possible and create a brighter future.

From this experience, which we poured our souls into, we have celebrated our successes and learnt from our failures, shared some incredible laughs and formed new or stronger bonds. Now it is time we look forward and see how you can play your own role in making Deep Learning Indaba $\mathbb{X}$ South Africa 2022 that much more special.